

## Course on Knowledge Graphs in the era of Large Language Models

### Session 2: Modelling OWL 2 Ontologies with Protégé

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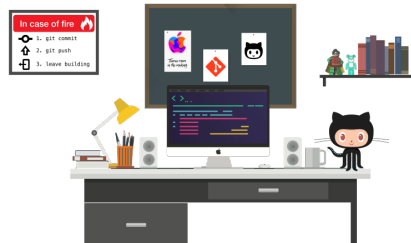
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# 1 Git Repository

Support codes for the laboratory sessions are available in *github*.

<https://github.com/city-knowledge-graphs/kg-course-valgrai>



## 2 First steps with ontology modelling

**Task 2.1:** Create an axiom (or axioms) for each of the following modelling tasks. Use DL or Manchester syntax to define the axioms (on paper):

1. All Mexican restaurants are restaurants.
2. Cities and countries are locations.
3. United Kingdom is a country.
4. An instance cannot be a pizza and a location at the same time.
5. Restaurants are located in cities.
6. Ernesto works for City, University of London
7. If a person teaches a module and that module belongs to a university then that person works for the university.
8. A vegan pizza contains only vegan ingredients.
9. A pizza with meat as ingredient is a meaty pizza.
10. A super veggie pizza contains at least 5 vegetarian ingredients.

## 3 The Pizza ontology

The pizza ontology and its tutorial are well-known resources in the Semantic Web community. They were developed for educational purposes by the University of Manchester. The tutorial have been recently updated by Michael DeBellis.

The pizza ontology and the tutorial are found at:

- <https://tinyurl.com/NewPizzaTutorialV3-2>
- <http://protege.stanford.edu/ontologies/pizza/pizza.owl>

In the following exercises we are going to explore the Pizza ontology and the ontology editor Protégé.<sup>1</sup>

**Task 2.2:** Open the pizza ontology in Protégé (*File menu and open from URL*). Take some time to browse the class hierarchy, the property hierarchies and the individuals and note how the ontology describes the domain of pizzas.

**Task 2.3:** Find `Margherita` and see how it is defined as a pizza with only cheese and tomato topping. Look at the definition of `VegetarianPizza`. Is a `MargheritaPizza` a vegetarian pizza? Why / why not?

**Task 2.4:** Find `hasIngredient`. What is the domain and range of this property? What are the subproperties of `hasIngredient`? What is the inverse property of `hasIngredient`? What property characteristics does `hasIngredient` have?

**Task 2.5:** Classify the ontology by choosing a reasoner (*e.g.*, `HermiT`) and then “classify/start reasoner” in the reasoner menu. In the entities/classes tab we can choose between the asserted classification and inferred classification (as a results of reasoning).

- In the “Inferred class hierarchy” two classes show up as subclasses of `owl:Nothing`. What does it mean for a class to be a subclass of `owl:Nothing`? Why these two classes appear as subclasses of `owl:Nothing`?
- Find `Margherita` in the inferred class hierarchy and see which classes are inferred as superclasses of `Margherita`.

**Task 2.6:** Add a new class `RomanoPizza` as a subclass of `NamedPizza`. Define `RomanoPizza` as a pizza with:

- `hasTopping some AnchoviesTopping`,
- `hasTopping some TomatoTopping and`
- `hasTopping some MozzarellaTopping`.

Classify the ontology. What superclasses are inferred as superclass of `RomanoPizza`? Why?

**Task 2.7:** Use the tutorial <https://tinyurl.com/NewPizzaTutorialV3-2> (also available in moodle). Chapters 1-3 give an overview of what we saw in the lecture. Chapters 4-8 give step by step instruction to create a version of the Pizza ontology.

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<sup>1</sup>Some additional documentation can be found here: <http://protegeproject.github.io/protege/>. The Protégé demo video from Lab 2 may also be helpful (check Lab 2 Task 2.5).

## 4 Pizza-Restaurants Dataset and Ontology

I have created an ontology based on the kaggle dataset about *Pizza Restaurants and the Pizza They Sell*.<sup>2</sup> I have uploaded in GitHub a dataset version that contains 500 data rows and the model ontology I created that covers the domain knowledge of the dataset, *i.e.*, the ontology conceptualizes/abstracts the information in the dataset.

**Task 2.8:** Explore the dataset and ontology. The idea of this task is to follow a reverse-engineering approach. Can you map the information in the dataset to the entities and axioms in the ontology? For example:

- Why do we have food ingredients?
- Why do we have an entity `Pizza bianca`?
- How are restaurant addresses captured?
- Why `City` and `Country` are represented as locations in the ontology?
- How types of restaurants are encoded in the ontology?
- Would you create a different ontology/conceptualization?

## 5 Some important considerations

There are some combinations of OWL 2 axioms that may make your ontology to fall outside OWL 2 DL, and thus falling into undecidability (*e.g.*, the reasoner will give an error in Protégé). For example, OWL 2 DL does not allow cardinality restrictions on transitive properties, but you can create this combination of axioms in Protégé.

**Task 2.9:** Read more about non-allowed combination of axioms in *Modeling in OWL 2 without Restrictions*: <https://arxiv.org/pdf/1212.2902.pdf>. You may encounter these issues when creating the ontology for the coursework Part 1.

## 6 Real-world ontologies

There are a large number of ontologies that are being currently applied in real-world solutions. Prominent examples are the ontologies in life sciences. For example, BioPortal (<https://bioportal.bioontology.org/>) contains  $\geq 1,000$  ontologies.

SNOMED CT ontology is one of the largest ( $\geq 300,000$ ) and most important ontologies. SNOMED CT vocabulary has been adopted by the NHS. There are also other smaller and more focused ontologies (*e.g.*, CMR-QA, an ontology I developed in the past, <https://bioportal.bioontology.org/ontologies/CMR-QA>).

Ontologies can also be found in many other domains (*e.g.*, Food: <https://foodon.org/>, Oil&Gas: <https://gitlab.com/logid/npd-factpages>).

**Task 2.10:** (Optional) Explore some of the above ontologies with Protégé.

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<sup>2</sup>Kaggle dataset about pizza restaurants: <https://www.kaggle.com/datafiniti/pizza-restaurants-and-the-pizza-they-sell>. This kaggle dataset is a subset of a dataset provided by Datafiniti's Business Database.

## 7 Solutions

### Task 2.1:

1. All Mexican restaurants are restaurants.

`MexicanRestaurant  $\sqsubseteq$  Restaurant`

2. Cities and countries are locations.

- Option 1:

`City  $\sqcup$  Country  $\sqsubseteq$  Location`

- Option 2 (preferred):

`City  $\sqsubseteq$  Location`

`Country  $\sqsubseteq$  Location`

3. United Kingdom is a country.

`Country(United_Kingdom)`

4. An instance cannot be a pizza and a location at the same time.

`Pizza  $\sqcap$  Location  $\sqsubseteq \perp$`

5. Restaurants are located in cities.

`Restaurant  $\sqsubseteq \exists \text{located\_in} . \text{City}$`

`Restaurant SubClassOf located_in some City`

6. Ernesto works for City, University of London

`works_for(ernesto, city_univ)`

7. If a person teaches a module and that module belongs to a university then that person works for the university.

`teaches  $\circ$  belongs_to  $\sqsubseteq$  works_for`

8. A vegan pizza contains only vegan ingredients.

`VeganPizza  $\sqsubseteq \forall \text{hasIngredient} . \text{VeganIngredient}$`

`VeganPizza SubclassOf hasIngredient only VeganIngredient`

9. A pizza with meat as ingredient is a meaty pizza.

`Pizza  $\sqcap \exists \text{hasIngredient} . \text{Meat} \sqsubseteq \text{MeatyPizza}$`

`Pizza and hasIngredient some Meat SubclassOf MeatyPizza`

10. A super veggie pizza contains at least 5 vegetarian ingredients.

`SuperVeggiePizza  $\sqsubseteq \geq 5 \text{hasIngredient} . \text{VegetarianIngredient}$`

`SuperVeggiePizza SubclassOf hasIngredient min 5 VegetarianIngredient`

**Task 2.3:**

Margherita is a vegetarian pizza as both cheese and tomato toppings are vegetarian toppings.

**Task 2.5:** The classes `CheesyVegetableTopping` and `IceCream` became subclasses of `owl:Nothing`. `CheesyVegetableTopping` has been declared as a subclass of `VegetableTopping` and `CheeseTopping` while these concepts are disjoint. The case of `IceCream` is explained in the lecture slides. Margherita has been classified as `CheesyPizza` and vegetarian pizza.

**Task 2.6:** `RomanoPizza` would be classified as a `CheesyPizza` (contains cheese), `NonVegetarianPizza` (contains anchovies), and `InterestingPizza` (it has at least 3 ingredients).

**Task 2.8:** Some insights about the dataset

- **Why do we have food ingredients?** The dataset describes pizzas sold by restaurants and there is a column about ingredients.
- **Why do we have an entity `Pizza bianca`?** One may want to model more pizzas than those in the dataset, but in this specific case, 'pizza bianca' does appear as one of the pizzas sold by a restaurant.
- **How are restaurant addresses captured?** Addresses are modelled as locations, and restaurants connect to the address via the property `locatedInAddress`.
- **Why `City` and `Country` are represented as locations in the ontology?** Check the lecture slides for more details.
- **How types of restaurants are encoded in the ontology?** Types of restaurant are encoded in a taxonomy with different levels.
- **Would you create a different ontology/conceptualization?** In many cases there are modelling choices that lead to alternative correct conceptualizations. So it is fine if you would model things in a different way.